

Executive Summary

Employee attrition is one of the major challenges faced by organizations because it directly impacts productivity, employee morale, operational efficiency, and hiring costs.

Understanding why employees leave an organization helps companies improve retention strategies and create a healthier work environment.

This project focuses on analyzing employee attrition patterns using Tableau dashboards and HR analytics techniques. The analysis was performed on an HR dataset containing 1,200 employee records with an **overall attrition rate of 25%**

The dashboard identifies both:

1. Where attrition is happening
2. Why attrition is happening

The analysis reveals that attrition is highly influenced by factors such as overtime, low job satisfaction, work-life imbalance, salary levels, and employee age groups. The project provides actionable insights that HR departments can use to improve employee retention and reduce turnover.

Problem Statement

Organizations often struggle with employee attrition because replacing employees requires additional recruitment, training, and operational costs. High attrition can also affect team performance, company culture, and long-term business growth.

The objective of this project is to:

- Analyze employee attrition patterns
- Identify departments and employee groups with higher attrition
- Understand the factors contributing to employee turnover
- Build an interactive Tableau dashboard for HR decision-making
- Provide business recommendations to improve employee retention

Project Objectives

The main objectives of this project are:

- To analyze employee attrition trends using HR data
- To identify which departments and job roles experience higher attrition
- To understand how salary, overtime, satisfaction levels, and work-life balance affect attrition
- To create an interactive and business-oriented Tableau dashboard
- To generate meaningful insights that support HR decision-making

Dataset Description

The dataset used in this project contains employee-related information including demographic details, job-related information, salary levels, satisfaction ratings, and attrition status.

Dataset Overview

- Total Employees: 1,200
- Attrition Count: 300
- Active Employees: 900
- Attrition Rate: 25%

Important Columns Used

- Age
- Age Group
- Gender
- Department
- Job Role
- Monthly Income
- Salary Slab
- Attrition
- Job Satisfaction
- Environment Satisfaction
- Work-Life Balance

- Overtime
- NumCompaniesWorked
- Education Field
- Marital Status

Data Preparation

The dataset was cleaned and transformed before visualization. Additional calculated fields and grouped categories were created to improve analysis quality.

Calculated/Derived Fields

- Age Group
- Salary Slab
- Attrition Percentage
- Active Employees
- Average Salary Metrics

Dashboard Overview

The dashboard was divided into two major analytical sections:

1. Where is Attrition Happening?

This section focuses on identifying the departments, employee groups, and categories where attrition is highest.

Charts Included:

- Attrition by Department
- Attrition by Job Role
- Attrition by Age Group
- Attrition by Gender

2. Why is Attrition Happening?

This section focuses on understanding the factors responsible for employee turnover.

Charts Included:

- Overtime vs Attrition
- Job Satisfaction vs Attrition
- Environment Satisfaction vs Attrition
- Work-Life Balance vs Attrition
- Employee Attrition Across Salary Levels

KPI Analysis

Total Employees



The organization contains a total workforce of 1,200 employees. This KPI provides the overall employee population used for analysis.

Attrition Count



A total of 300 employees left the organization. This KPI helps measure the scale of employee turnover.

Attrition Rate



The overall attrition rate is 25%, indicating that one out of every four employees left the company.

Active Employees



There are currently 900 active employees in the organization after excluding attrition cases.

Average Monthly Salary



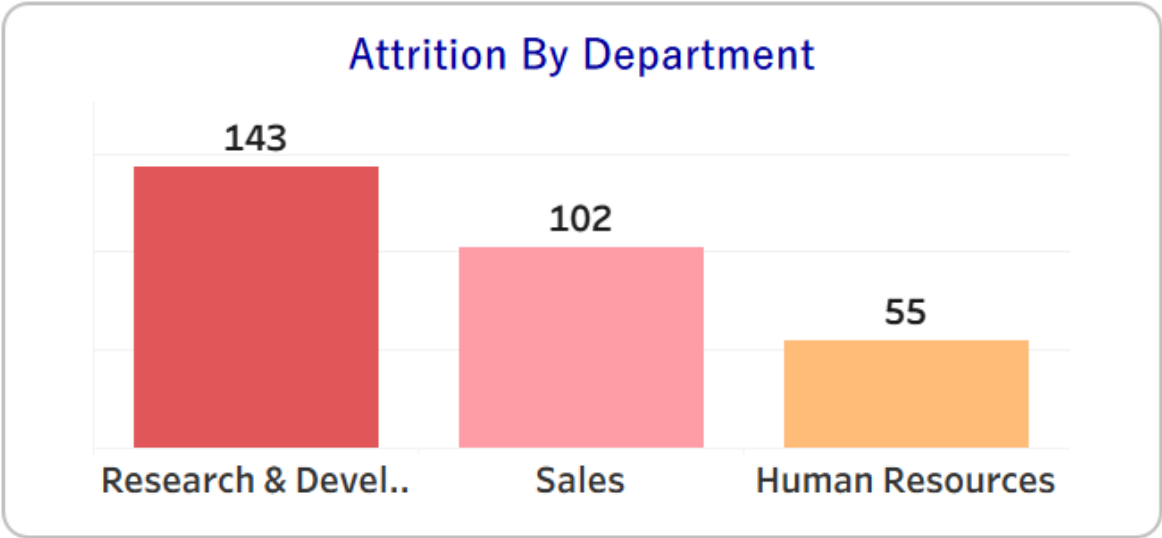
The average monthly salary across employees is 83,619. This KPI helps compare compensation patterns against attrition trends.

Dashboard Analysis & Insights

Section 1: Where is Attrition Happening?

Attrition by Department

The Research & Development department shows the highest attrition count compared to Sales and Human Resources.



Key Insights

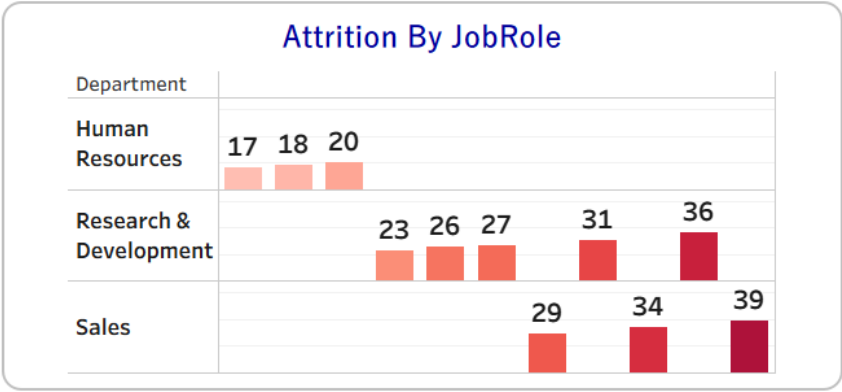
- Research & Development has the highest employee turnover.
- Sales department also experiences significant attrition.
- Human Resources has comparatively lower attrition.

Business Interpretation

The high attrition in Research & Development may indicate workload pressure, limited growth opportunities, or higher competition for skilled employees in the market.

Attrition by Job Role

The dashboard analyzes attrition across different job roles within each department.



Key Insights

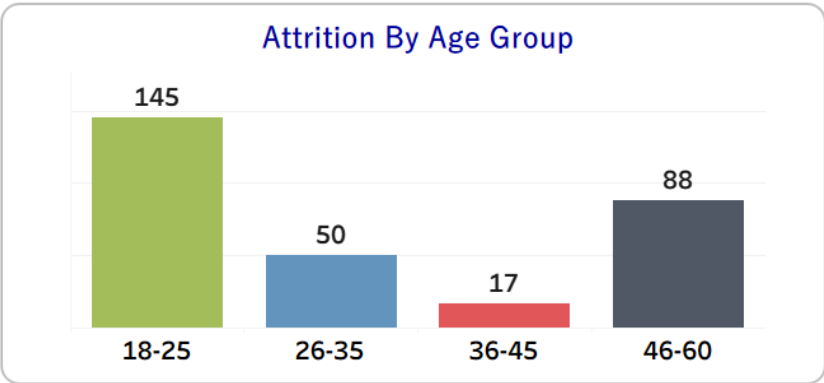
- Certain job roles within Sales and Research & Development contribute heavily to attrition.
- Technical and operational roles show comparatively higher employee exits.

Business Interpretation

Employees in demanding operational or technical positions may experience higher stress levels, burnout, or better opportunities outside the organization.

Attrition by Age Group

The age group analysis shows that younger employees contribute the highest attrition.



Key Insights

- Employees aged 18–25 show the highest attrition.
- Employees aged 46–60 also show noticeable attrition.

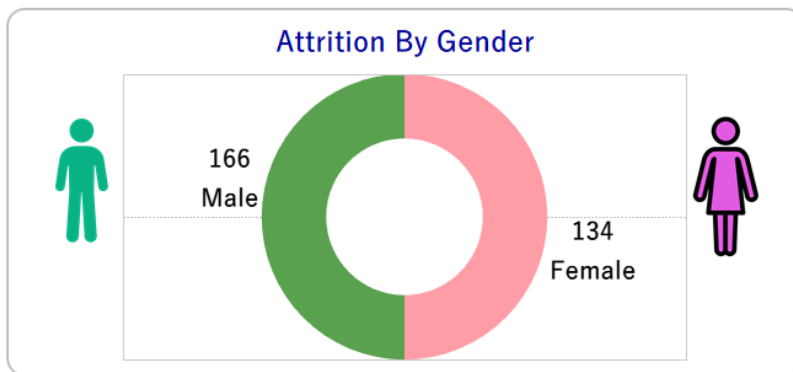
- Employees between 36–45 show the lowest attrition.

Business Interpretation

Younger employees are more likely to switch jobs for better opportunities, salary growth, or career advancement. Senior employees may leave due to retirement planning or career transitions.

Attrition by Gender

The gender analysis compares attrition distribution between male and female employees.



Key Insights

- Male employees show slightly higher attrition compared to female employees.
- Attrition is distributed across both genders relatively evenly.

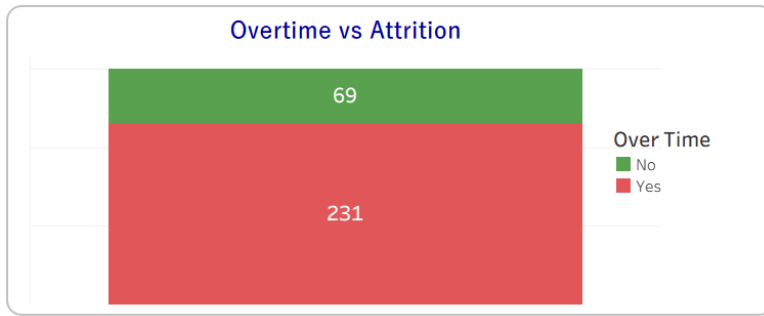
Business Interpretation

The organization may need gender-neutral retention strategies while focusing more on role-specific and operational factors.

Section 2: Why is Attrition Happening?

Overtime vs Attrition

This chart analyzes the relationship between overtime work and employee attrition.



Key Insights

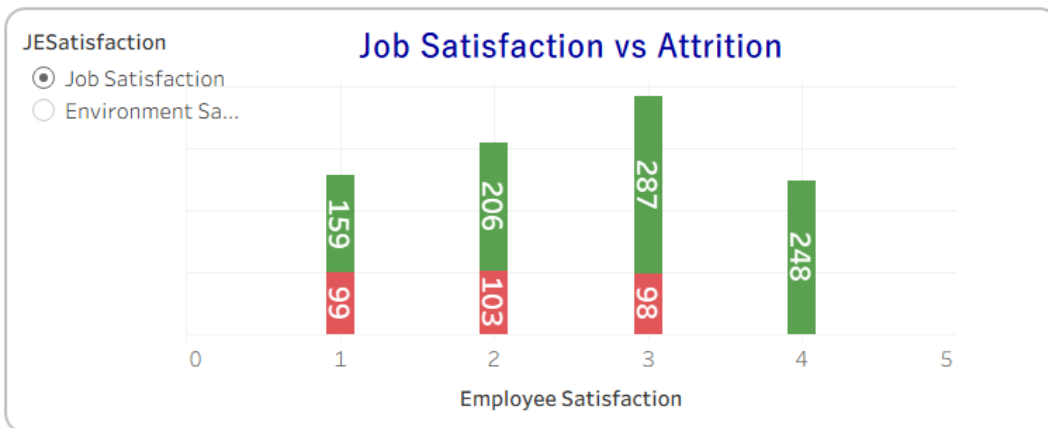
- Employees working overtime contribute significantly to attrition.
- A large percentage of employees leaving the organization had overtime responsibilities.

Business Interpretation

Excessive overtime can lead to stress, burnout, fatigue, and poor work-life balance, ultimately increasing employee turnover.

Job Satisfaction vs Attrition

This analysis measures how employee satisfaction impacts attrition.



Key Insights

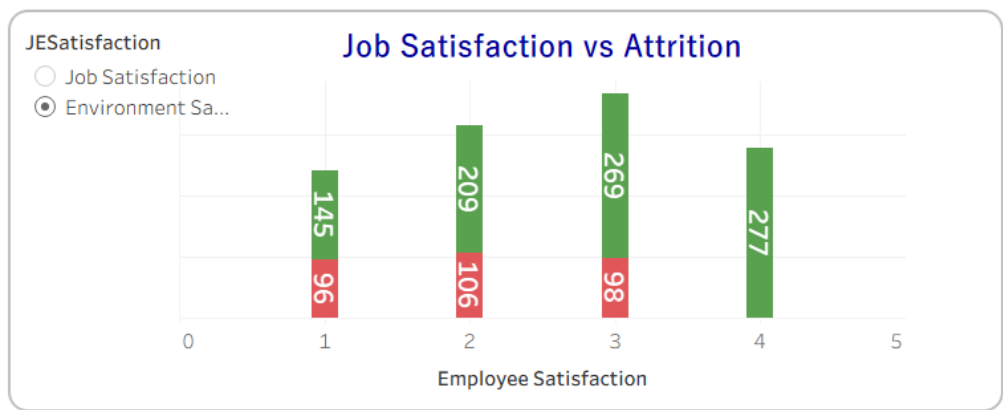
- Lower job satisfaction levels are associated with higher attrition.
- Employees with higher satisfaction levels show better retention.

Business Interpretation

Employees who are dissatisfied with their roles, management, work environment, or career growth are more likely to leave the organization.

Environment Satisfaction vs Attrition

This section analyzes employee satisfaction with the workplace environment.



Key Insights

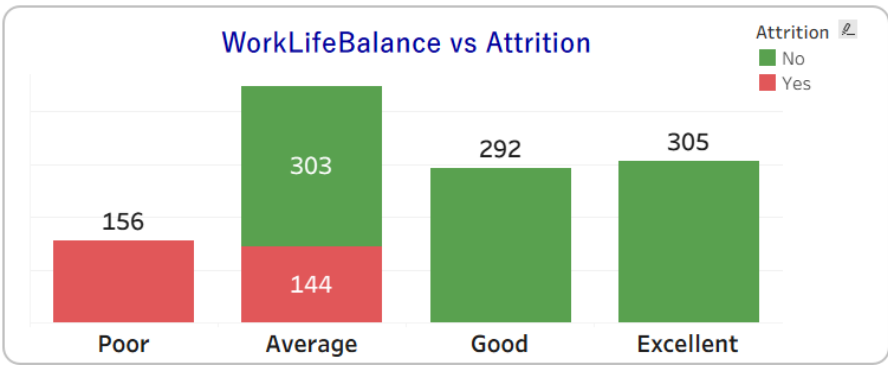
- Lower environment satisfaction scores correlate with higher attrition.
- Positive workplace environments improve employee retention.

Business Interpretation

A supportive and engaging work environment plays an important role in retaining employees and improving morale.

Work-Life Balance vs Attrition

The chart evaluates how work-life balance impacts employee turnover.



Key Insights

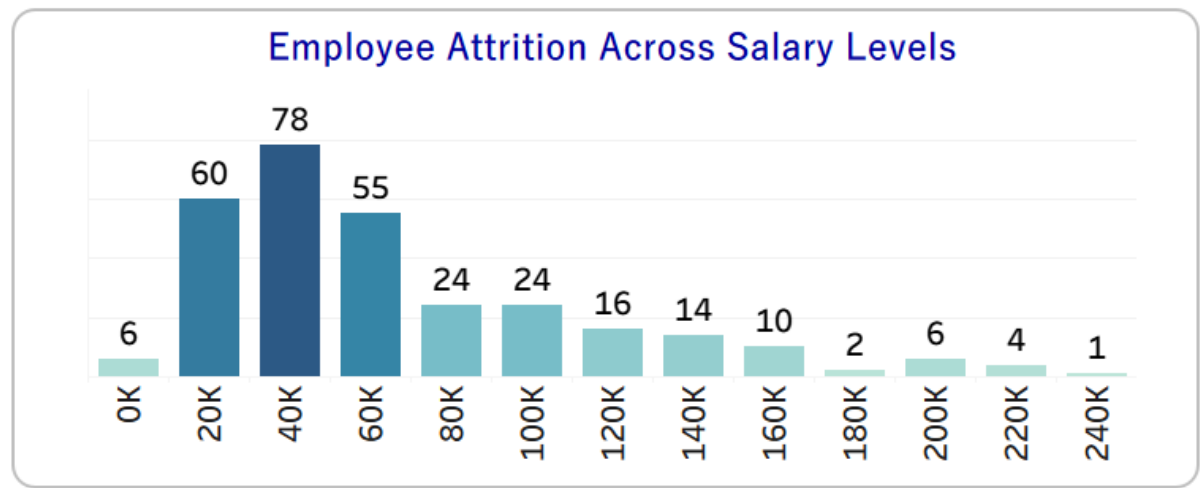
- Employees with poor work-life balance show the highest attrition.
- Employees with good or excellent work-life balance show better retention.

Business Interpretation

Organizations that fail to maintain healthy work-life balance policies may experience higher employee dissatisfaction and resignations.

Employee Attrition Across Salary Levels

This analysis identifies attrition trends across different salary slabs.



Key Insights

- Lower salary levels experience higher attrition.
- Attrition gradually decreases as salary levels increase.

Business Interpretation

Compensation plays a major role in employee retention. Employees with lower salaries are more likely to leave for better financial opportunities.

Major Findings

The analysis identified several major drivers of employee attrition:

- Overtime strongly contributes to employee turnover.
- Younger employees are more likely to leave the organization.
- Lower salary slabs experience higher attrition.
- Poor work-life balance increases resignation rates.
- Job dissatisfaction is closely linked to attrition.
- Research & Development department experiences the highest attrition.

Recommendations

Based on the analysis, the following recommendations are suggested:

1. Improve Work-Life Balance

- Reduce excessive overtime
- Introduce flexible working policies
- Encourage healthier work schedules

2. Increase Employee Engagement

- Conduct regular employee feedback surveys
- Improve communication between management and employees
- Introduce employee recognition programs

3. Review Compensation Structure

- Improve salary competitiveness for lower salary bands
- Provide performance-based incentives
- Introduce retention bonuses for critical roles

4. Strengthen Career Development

- Create clear career growth opportunities
- Offer learning and development programs
- Improve internal promotion pathways

5. Focus on Younger Employees

- Improve onboarding experience
- Provide mentorship programs
- Create long-term career planning initiatives

Conclusion

This HR Analytics project successfully identified the major patterns and factors responsible for employee attrition using Tableau dashboards and HR data analysis techniques.

The project demonstrates how business intelligence and data visualization tools can help organizations understand employee behavior and improve strategic HR decision-making.

By analyzing attrition across departments, age groups, salary levels, work-life balance, and satisfaction metrics, the dashboard provides meaningful business insights that can support employee retention strategies.